

Extreme Weather Events (hurricanes, floods, Droughts): Intensification and increased frequency leading to environmental & societal impacts.



# Analysing the Issue:

how hurricanes, floods, and droughts are becoming more frequent and severe ?

Rising Global Temperatures, Higher sea levels worsen storm surges during hurricanes and cyclones, increasing flood risks in coastal areas, Changing Rainfall Patterns etc.

Influence of climate change and human activities in driving these changes.

Rising global temperatures → more energy in atmosphere & oceans → stronger hurricanes.

Warmer air holds more moisture → heavier rainfall & floods. Increased heat → more frequent & severe droughts.

Deforestation & urbanization → loss of natural flood protection. Greenhouse gas emissions → accelerate global warming.

Overuse of groundwater & poor water management → worsen drought conditions.

Impacts on ecosystems, agriculture, economy, and human life.

Ecosystems – Habitat loss, species extinction, biodiversity decline.

Agriculture – Crop failures, soil infertility, water scarcity.

Economy – Infrastructure damage, high recovery costs, job losses.

Human life – Displacement, health risks, loss of homes & security.

# How AI can be implemented to tackle extreme weather and climate change

Early Warning Systems – AI predicts hurricanes, floods, and droughts using satellite and climate data.

Disaster Response – Guides evacuation, resource allocation, and damage assessment with real-time analysis.

Climate Modelling – Improves accuracy of long-term climate predictions and identifies high-risk regions.

Agriculture – Provides farmers with smart irrigation, crop monitoring, and pest/disease detection.

Water Management – Optimizes reservoir use, monitors river levels, and prevents water wastage.

Energy Efficiency – Supports renewable energy integration and reduces carbon emissions.

Urban Planning – Helps design climate-resilient infrastructure and flood-proof cities.

# Real-Life Examples of AI Tackling Extreme Weather & Climate Change:

Google Flood Forecasting Initiative – Uses AI + satellite data to predict floods and provide early warnings to vulnerable regions (like India and Bangladesh).

IBM's Green Horizon Project – Uses AI to forecast air pollution and optimize renewable energy use.

Microsoft AI for Earth – Supports projects that use AI to monitor agriculture, water, and biodiversity.

European Centre for Medium-Range Weather Forecasts (ECMWF) – Uses AI-driven models for more accurate hurricane and climate predictions. [Can be found in windy.com]

AccuWeather – Uses AI-powered models for highly accurate hurricane forecasts, hyper-local flood/drought alerts, and ClimateReady™ tools to help governments, businesses, and communities prepare for long-term climate risks.

# Conclusion:

Extreme weather events like hurricanes, floods, and droughts are increasing in frequency and intensity due to climate change and human activities.

These events severely impact ecosystems, agriculture, economy, and human life, causing habitat loss, crop failures, infrastructure damage, and displacement

AI plays a crucial role in addressing these challenges through:

Accurate predictions and early warning systems.

Disaster management and planning of evacuation/resources

Climate modeling and risk assessment for long-term planning.

Precision agriculture and optimized water/energy management.

Practical Applications of AI in Tackling Climate Challenges:

Real-life examples, including AccuWeather, Google Flood Forecasting, IBM Green Horizon, and Microsoft AI for Earth, demonstrate AI's effectiveness in improving forecasting, preparedness, and resilience.

Integrating AI-driven solutions with climate awareness helps minimize damage, protect communities, and build sustainable resilience against extreme weather events.